

Finite Elasticity

Arrangements

- Twelve lectures on seismology and Earth structure
- Two linked problem sets – solutions online immediately
- Supervisions in the weeks of **17th November** and **1st December** – sign up information to come
- Change in lecture style – **fewer topics, more detail**
- Lecture content builds directly on each other

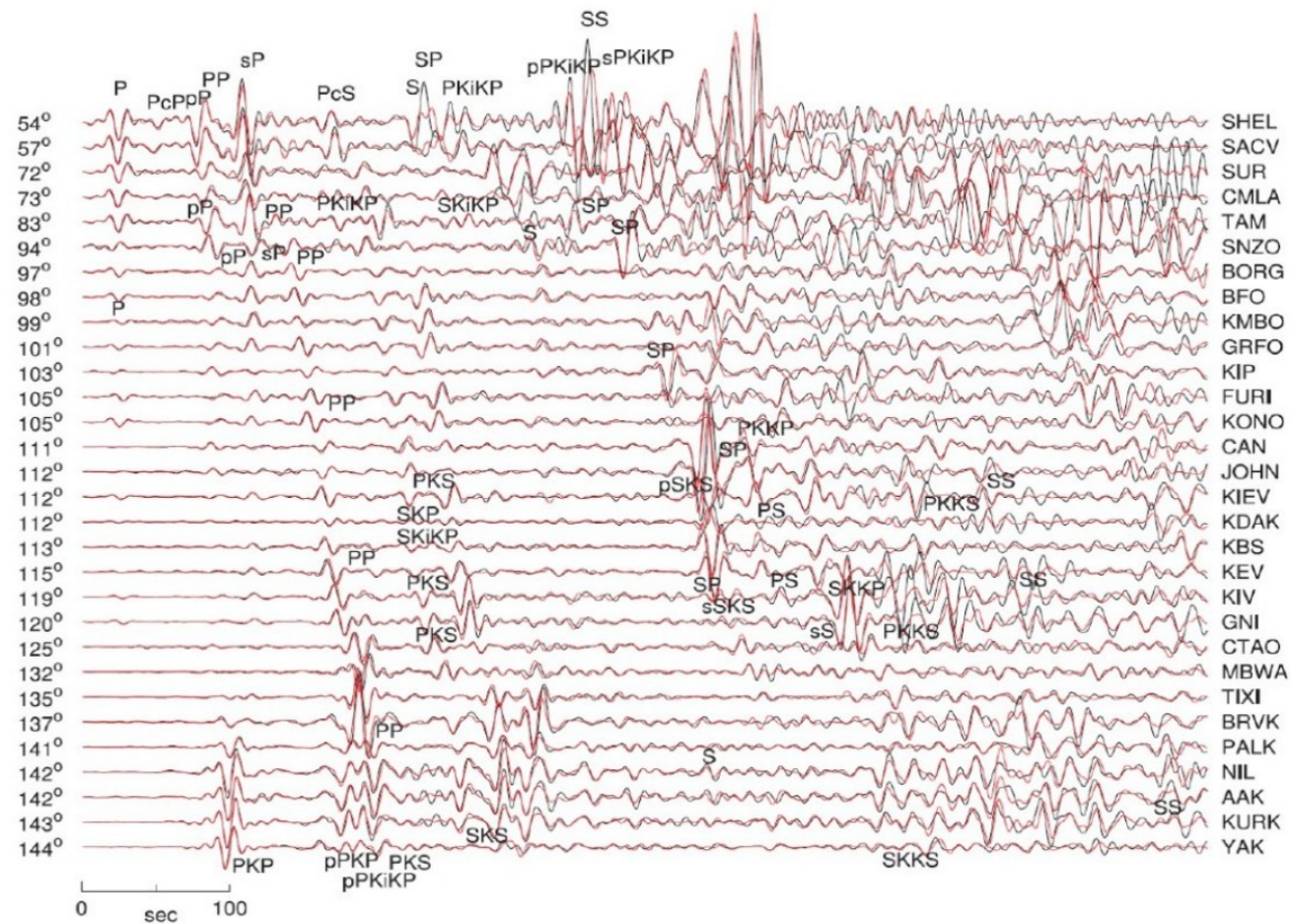
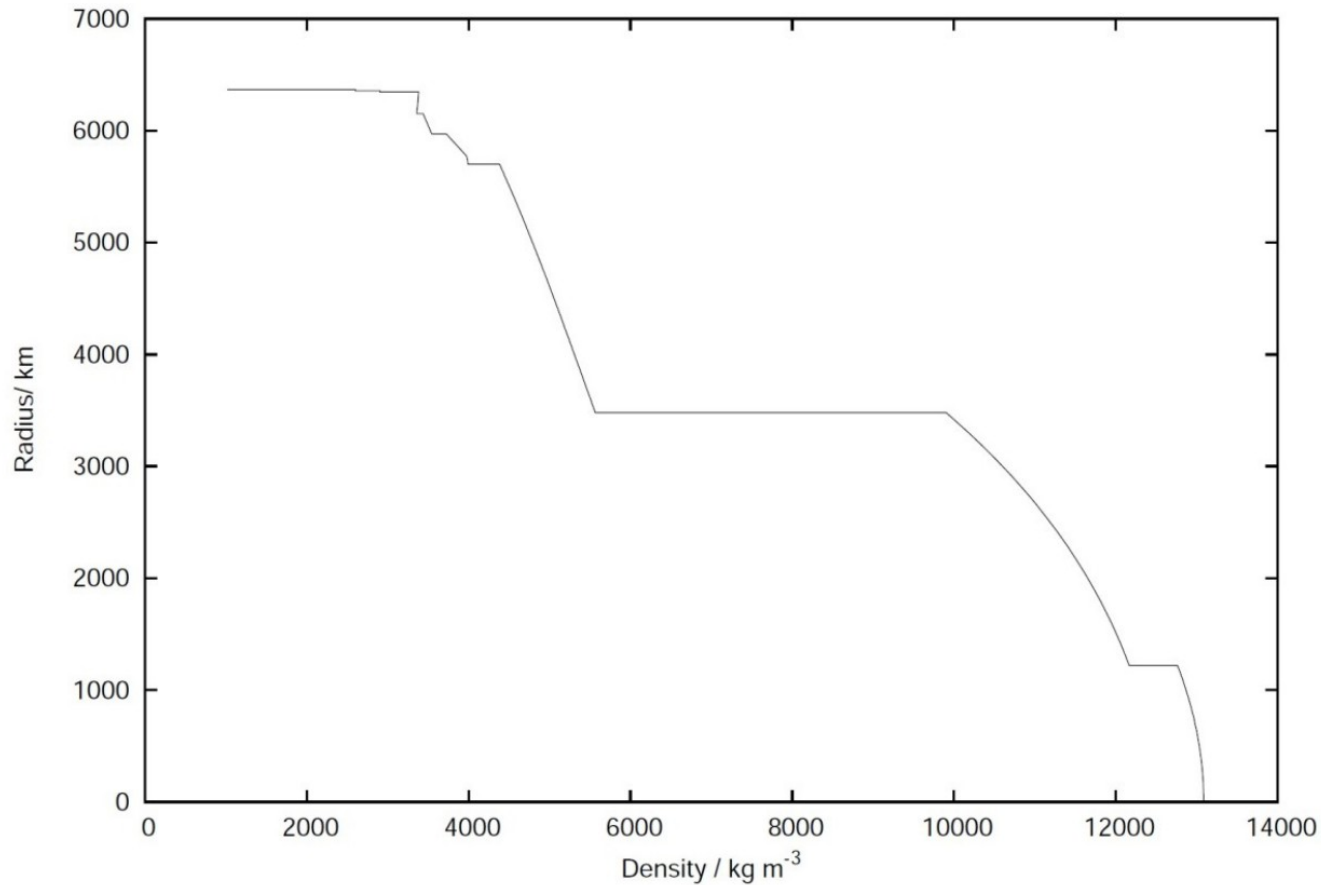


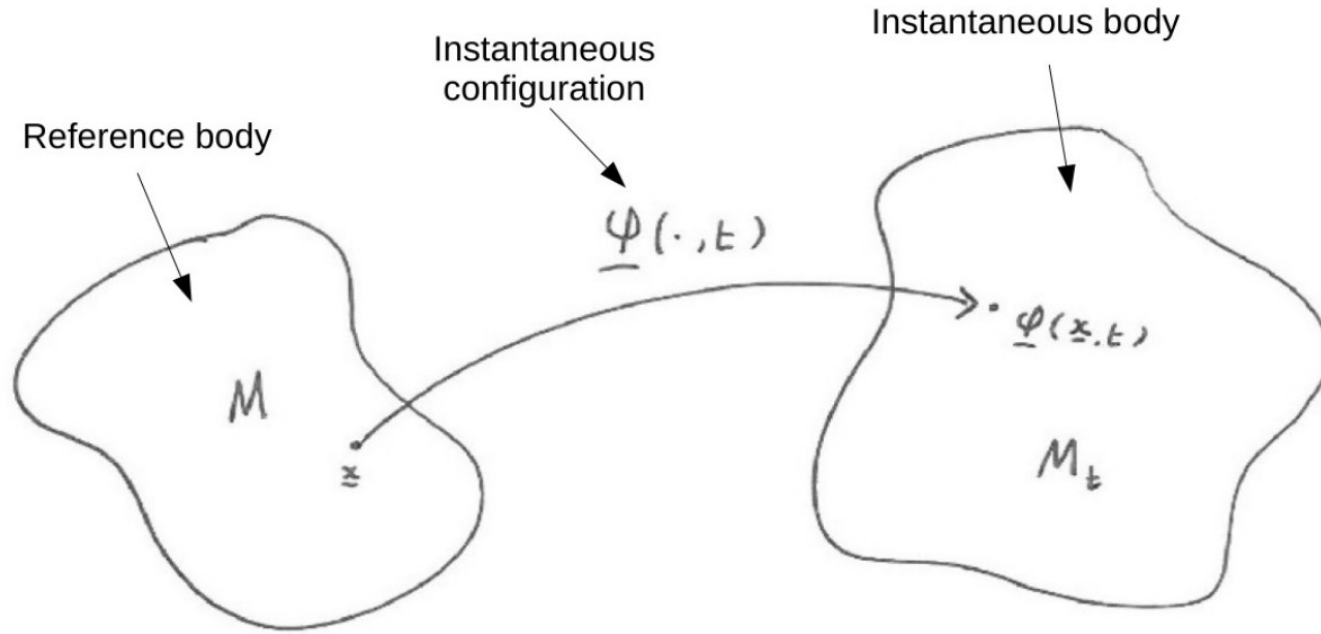
Figure 7. Vertical component record section comparing data (black) and SEM synthetics (red) for the 2008 September 3, $M_w = 6.3$ Santiago del Estero, Argentina earthquake, which occurred at a depth of 571 km. The records are aligned on the P wave, plotted as a function of epicentral distance, and bandpass filtered between 17 and 60 s. Major seismicological body wave arrivals are labelled. Epicentral distance is plotted to the left of each set of traces, and FDSN station identification codes are plotted to the right.



Model of Earth's interior density structure. Believed to be accurate enough to conclude that the outer core is less dense than would be expected for an iron-nickel alloy. Suggests there is a “light element” in the core. Tells us something about planetary formation and differentiation.

Kinematics

Motion of a body



$$\varphi(\mathbf{x}, t_0) = \mathbf{x}$$

Referential vs spatial descriptions

- **Referential quantities** are functions of position within the **reference body**
- **Spatial quantities** are functions of position within **physical space**
- In either case, functions might also depend on time
- In fluid mechanics, spatial (or **Eulerian**) descriptions most common
- In solid mechanics, referential (or **Lagrangian**) descriptions most common
- These conventions follow ultimately from the **constitutive relations** involved

Referential velocity

Referential velocity is the velocity of fixed particles:

$$v_i(\mathbf{x}, t) = \frac{\partial \varphi_i}{\partial t}(\mathbf{x}, t)$$

Can be related to the **spatial velocity** using the motion

We will always use the referential velocity, and typically just call it *the* velocity

Deformation gradient

First-order Taylor expansion of the motion about a fixed point:

$$\varphi_i(\mathbf{x} + \delta\mathbf{x}, t) = \varphi_i(\mathbf{x}, t) + F_{ij}(\mathbf{x}, t)\delta x_j + \dots$$

Deformation gradient has components:

$$F_{ij} = \frac{\partial \varphi_i}{\partial x_j}$$

Related to the **strain tensor** – more details in the next lecture

Jacobian

The **Jacobian** of the motion is defined by:

$$J(\mathbf{x}, t) = \det \mathbf{F}(\mathbf{x}, t).$$

Must be non-zero if the instantaneous configurations are invertible. We can assume it is **always positive** so that the motion is **orientation preserving**

Can be used to transform spatial integrations to the reference body:

$$\int_{M_t} f(\mathbf{y}) \, \mathrm{d}^3 \mathbf{y} = \int_M f[\varphi(\mathbf{x}, t)] J(\mathbf{x}, t) \, \mathrm{d}^3 \mathbf{x}$$

Conservation of mass

Integral conservation laws

For an arbitrary sub-body, we must have:

$$\frac{d}{dt} \int_{U_t} \varrho(\mathbf{y}, t) d^3\mathbf{y} = 0.$$

Transforming the integral to the reference body, we find:

$$\frac{d}{dt} \int_U J(\mathbf{x}, t) \varrho[\boldsymbol{\varphi}(\mathbf{x}, t), t] d^3\mathbf{x} = 0.$$

Local conservation laws

Because the sub-body was arbitrary, we must have:

$$\frac{\partial}{\partial t} \{J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t]\} = 0.$$

This leads us to define the **referential density**:

$$\rho(\mathbf{x}) = J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t]$$

Note slight notation change between printed handouts

Referential density is just the density within the reference configuration

Kinetic and potential energies

Kinetic energy

Kinetic energy density is defined by analogy with particle mechanics:

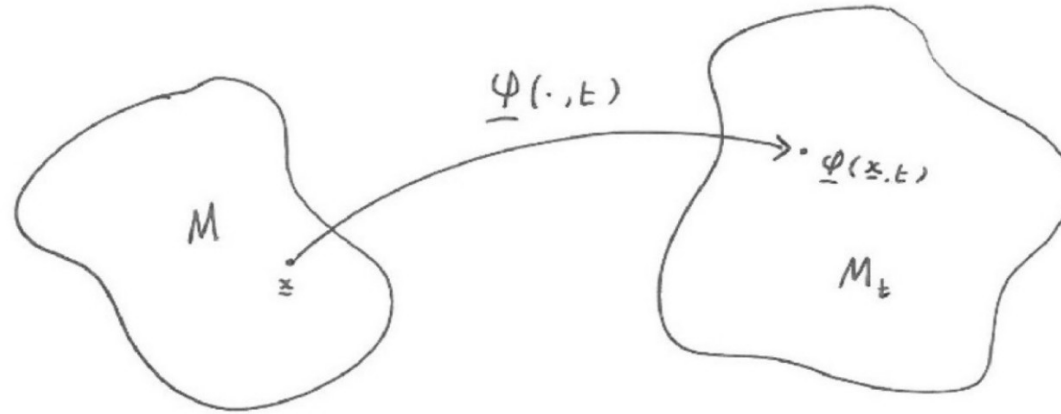
$$\frac{1}{2}\rho(\mathbf{x})||\mathbf{v}(\mathbf{x}, t)||^2,$$

Note this definition is per unit volume in the reference body

Kinetic energy of the body at an instance in time is given by:

$$\mathcal{T}(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) ||\mathbf{v}(\mathbf{x}, t)||^2 \mathrm{d}^3\mathbf{x}.$$

Hyperelastic bodies



Elastic bodies have a **preferred shape and/or volume**, with work being necessary to deform them away from this preferred state

If the work required depends only on the **final configuration**, there is a well-defined **potential energy**. Such bodies are said to be **hyperelastic**

Elastic potential energy

We assume that the potential energy takes the form:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x},$$

This choice *defines* a **simple hyperelastic** body

The term **simple** means that the potential energy density at a point depends only on the motion and its first spatial derivative at the same point

It's possible to consider dependence on higher-order spatial derivatives at a point, or a non-local form for the potential energy density, but this is not obviously needed within real materials

The form of the potential energy can be simplified using symmetry arguments – this is an application of the **principle of material frame indifference**

Time-translation invariance

If two motions differ only by a time-translation, we require that:

$$\mathcal{V}'(t) = \mathcal{V}(t - \tau).$$

This condition is met if and only if the potential energy density simplifies:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \varphi(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x}.$$

No explicit time-dependence of the potential energy density

Super-imposed rigid body motions

Two motions differing by a **rigid body motion** can be related by:

$$\varphi'(\mathbf{x}, t) = \mathbf{a}(t) + \mathbf{Q}(t)\varphi(\mathbf{x}, t)$$

We require the motions to have the same potential energy

A sufficient condition is that:

$$W[\mathbf{x}, \varphi(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{a}(t) + \mathbf{Q}(t)\varphi(\mathbf{x}, t), \mathbf{Q}(t)\mathbf{F}(\mathbf{x}, t)]$$

This can also be shown to be a necessary condition

Super-imposed rigid body motions

Focusing first on spatial translations, we conclude that

$$W[\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{a}(t) + \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)].$$

No explicit dependence on the **particle's position in space**

Potential energy now reduced to:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x}.$$

$$W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{Q}(t)\mathbf{F}(\mathbf{x}, t)],$$

The potential energy density is conventionally called the **strain energy function**

Hamilton's principle

Action and Lagrangian

Variational arguments covered from **first-principles** and in full detail in the handout – please ask if you need any additional help

Action for the for a motion is defined by:

$$\mathcal{S}(\varphi) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x} dt,$$

Where the **Lagrangian** takes the conventional form:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, \mathbf{F})$$

To each possible motion of the body, the action returns a single real number – it defines a **functional** of the body's motion

Hamilton's principle

Functional derivative of the action must vanish at the true motion:

$$\delta S = \lim_{h \rightarrow 0} \frac{1}{h} [\mathcal{S}(\varphi + h \delta\varphi) - \mathcal{S}(\varphi)]$$

so long as the variations to the motion have **fixed-endpoints**

$$\delta\varphi(\mathbf{x}, 0) = \delta\varphi(\mathbf{x}, T) = 0.$$

This is **Hamilton's principle**, and it is **equivalent to Newton's laws of motion** for systems in which a potential energy is defined

Variation of the action

To first-order in the perturbed motion, we have:

$$\delta \mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial \mathcal{L}}{\partial \varphi_i} \delta \varphi_i + \frac{\partial \mathcal{L}}{\partial v_i} \delta v_i + \frac{\partial \mathcal{L}}{\partial F_{ij}} \delta F_{ij} \right\} d^3 \mathbf{x} dt,$$

This can be simplified by relating variations in velocity and deformation gradient to the underlying motion:

$$\begin{aligned} \delta v_i &= \frac{\partial \delta \varphi_i}{\partial t} \\ \delta F_{ij} &= \frac{\partial \delta \varphi_i}{\partial x_j} \end{aligned}$$

Variation of the action

Using the product rule, we see that:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial v_i} \delta v_i &= \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \delta \varphi_i \right) - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \right) \delta \varphi_i \\ \frac{\partial \mathcal{L}}{\partial F_{ij}} \delta F_{ij} &= \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \delta \varphi_i \right) - \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \right) \delta \varphi_i,\end{aligned}$$

We can then integrate by parts in time, and use the divergence theorem to simplify the expression:

$$\delta \mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial \mathcal{L}}{\partial \varphi_i} \delta \varphi_i + \frac{\partial \mathcal{L}}{\partial v_i} \delta v_i + \frac{\partial \mathcal{L}}{\partial F_{ij}} \delta F_{ij} \right\} d^3 \mathbf{x} dt,$$

Variation of the action

Following this effort, the first order change in the action is:

$$\delta\mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial\mathcal{L}}{\partial\varphi_i} - \frac{\partial}{\partial t} \left(\frac{\partial\mathcal{L}}{\partial v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\partial\mathcal{L}}{\partial F_{ij}} \right) \right\} \delta\varphi_i \, d^3\mathbf{x} \, dt \\ + \int_0^T \int_{\partial M} \frac{\partial\mathcal{L}}{\partial F_{ij}} \hat{n}_j \delta\varphi_i \, dS \, dt$$

Fixed-endpoint conditions have been used

We need the above expression to vanish for all such variations of the motion

Euler-Lagrange equations

From Hamilton's principle, we have obtained the **Euler-Lagrange equation**:

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \right) = 0,$$

along with the **natural boundary condition**

$$\frac{\partial \mathcal{L}}{\partial F_{ij}} \hat{n}_j = 0,$$

These equations govern the motion of a simple hyperelastic body

First Piola-Kirchhoff stress

For our Lagrangian, we see that

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} = 0, \quad \frac{\partial \mathcal{L}}{\partial v_i} = \rho v_i \quad \frac{\partial \mathcal{L}}{\partial F_{ij}} = -\frac{\partial W}{\partial F_{ij}}.$$

We define the **first Piola-Kirchhoff stress tensor** to be

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)],$$

Conservation of linear momentum

Euler-Lagrange equations can now be written:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0,$$

subject to the boundary conditions

$$T_{ij} \hat{n}_j = 0.$$

In the third problem set, you will show these equations are equivalent to the **conservation of linear momentum**

You will also learn about the physical meaning of this new stress tensor